

## No. of Functional Features Included in the Solution

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 01 July 2026                    |
| <b>Team ID</b>       | SWTID-2026-8998                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S.No | Feature Name                    | Feature Description                                                                                    | Module / Component | Status (Done / In Progress / Pending) | Marks Contribution |
|------|---------------------------------|--------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------|--------------------|
| 1    | Applicant Data Entry            | Collects applicant personal, financial, and employment details through a web form.                     | Frontend           | Done                                  | High               |
| 2    | Data Preprocessing              | Cleans, transforms, and prepares applicant data for prediction.                                        | Backend            | Done                                  | High               |
| 3    | Credit Card Approval Prediction | Predicts whether the applicant is eligible using the trained ML model.                                 | Backend            | Done                                  | High               |
| 4    | Model Comparison                | Compares Logistic Regression, Decision Tree, Random Forest, and XGBoost models to select the best one. | Machine Learning   | Done                                  | Medium             |
| 5    | Result Display                  | Displays the approval or rejection prediction to the user.                                             | Frontend           | Done                                  | High               |
| 6    | Flask Web Application           | Integrates the frontend with the machine learning model for real-time prediction.                      | Backend            | Done                                  | High               |
| 7    | IBM Watson Deployment           | Deploys the trained model for cloud-based prediction.                                                  | Cloud Deployment   | Done                                  | Medium             |
| 8    | Input Validation                | Validates user inputs before sending data for prediction.                                              | Backend            | Done                                  | Medium             |

**Feature Summary:**

| Metric                             | Count / Value |
|------------------------------------|---------------|
| Total Features Planned             | 8             |
| Total Features Implemented         | 8             |
| Core / Must-Have Features          | 6             |
| Additional / Nice-to-Have Features | 2             |
| Features Tested & Verified         | 8             |

**Feature Category Breakdown:**

| S.No | Category                  | Features in Category | Example Features                                        |
|------|---------------------------|----------------------|---------------------------------------------------------|
| 1    | User Interface (UI)       | 2                    | Applicant Data Entry, Result Display                    |
| 2    | Backend / Logic           | 3                    | Flask Application, Data Preprocessing, Input Validation |
| 3    | Database / Storage        | 1                    | Credit Card Dataset, Trained Model (.pkl)               |
| 4    | API / Integration         | 1                    | IBM Watson Machine Learning Deployment                  |
| 5    | Security / Authentication | 1                    | Input Validation and Error Handling                     |